

ELHOVO, Bulgaria

WMO: 156420

Lat: 42.183N

Lon: 26.567E

Elev: 139

StdP: 99.67

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-7.5	-13.4	1.2	-8.8	-10.9	1.5	-5.9	17.2	2.6	16.6	2.4	1.8	0	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	34.4	23.5	32.7	22.5	31.2	21.7	25.4	32.2	24.1	30.4	22.9	29.0	3.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.4	30.4	22.1	17.0	28.8	20.8	15.8	26.7	78.9	32.2	73.4	30.4	68.5	29.1	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	9.3	7.4	-13.8	37.5	3.7	2.5	-16.5	39.3	-18.7	40.8	-20.8	42.2	-23.6	44.1		
(5)			-14.2	25.7	3.7	2.8	-16.8	27.6	-19.0	29.3	-21.0	30.8	-23.7	32.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	1.6	3.6	7.3	12.0	16.8	20.9	23.4	23.5	19.0	13.5	8.0	2.8		
	DBStd	8.62	4.84	4.84	4.26	3.55	3.16	2.77	2.42	2.49	3.22	4.01	4.46	5.37		
	HDD10.0	907	261	182	107	21	1	0	0	0	14	91	230			
	HDD18.3	2517	518	412	342	192	66	8	1	30	156	311	481			
	CDD10.0	1910	2	4	23	79	212	327	415	418	269	122	31	8		
	CDD18.3	478	0	0	0	1	18	86	157	161	49	6	1	0		
	CDH23.3	5403	0	0	1	19	274	969	1791	1783	511	55	1	0		
	CDH26.7	2178	0	0	0	2	53	347	800	814	153	8	0	0		
(14) Wind	WSAvg	2.0	1.9	2.6	2.9	2.4	2.0	1.7	1.9	1.8	1.8	1.9	1.7	2.0		
(15) Precipitation	PrecAvg	555	42	41	40	45	49	47	44	28	50	57	54	58		
	PrecMax	850	91	140	85	96	122	119	166	85	153	149	144	134		
	PrecMin	364	2	2	4	8	19	2	1	0	2	3	0	14		
	PrecStd	117	27	32	21	23	30	28	38	23	42	38	37	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	17.7	21.7	25.9	30.4	34.1	37.6	36.8	32.6	27.6	21.9	18.1		
		MCWB	11.9	12.1	14.0	18.1	20.5	23.6	25.2	24.6	22.0	18.6	15.5	15.8		
	2%	DB	12.7	15.4	19.2	22.9	28.2	32.0	34.6	34.6	30.2	24.6	19.1	14.8		
		MCWB	10.1	10.7	13.3	15.7	19.8	22.4	23.4	23.8	20.7	18.0	14.9	12.7		
	5%	DB	10.5	13.3	17.0	20.9	26.3	30.2	32.6	32.7	28.3	22.5	16.9	12.8		
		MCWB	8.5	9.6	11.8	14.5	18.6	21.4	22.4	22.7	19.5	16.7	13.7	11.0		
	10%	DB	8.3	11.0	14.8	19.1	24.6	28.6	31.0	31.0	26.3	20.3	14.7	10.7		
		MCWB	7.0	8.3	10.6	13.5	17.7	20.5	21.6	21.7	18.5	15.6	12.3	9.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.3	15.3	18.2	22.5	26.1	27.6	27.3	23.6	20.3	16.9	15.9		
		MCDB	14.4	16.5	19.8	24.2	26.9	32.4	35.0	34.9	30.0	25.1	19.8	17.7		
	2%	WB	10.6	11.5	13.5	16.3	20.8	24.2	25.4	25.4	21.6	18.8	15.4	13.2		
		MCDB	12.2	14.1	18.0	21.8	26.4	30.0	32.1	32.5	27.8	22.8	18.2	14.6		
	5%	WB	8.7	9.9	12.2	15.0	19.5	22.6	24.0	24.1	20.5	17.6	14.1	11.2		
		MCDB	10.1	12.6	16.1	20.1	25.2	28.3	30.0	30.7	26.5	21.3	16.5	12.5		
	10%	WB	7.1	8.4	11.0	13.9	18.2	21.3	22.8	22.7	19.3	16.3	12.6	9.5		
		MCDB	8.3	10.8	14.6	18.2	23.5	26.9	29.0	29.0	24.7	19.7	14.5	10.7		
(35) Mean Daily Temperature Range	MDBR		6.8	8.8	10.1	11.3	13.1	13.5	14.0	14.2	12.5	10.2	8.5	6.7		
	5% DB	MCDBR	10.0	12.6	13.7	14.4	16.3	16.3	17.0	16.9	16.0	13.3	10.8	8.0		
		MCWBR	7.4	8.3	8.2	8.2	8.6	7.6	7.7	7.5	7.5	7.1	6.5	6.0		
	5% WB	MCDBR	9.1	11.3	12.4	13.2	14.8	15.5	15.9	15.9	14.5	12.0	9.7	7.3		
		MCWBR	7.2	8.0	7.9	8.0	8.6	8.8	9.0	8.5	7.7	7.3	6.2	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.378	0.419	0.469	0.443	0.444	0.447	0.455	0.433	0.400	0.373	0.351		
	taud		2.325	2.239	2.143	2.013	2.165	2.202	2.206	2.181	2.214	2.286	2.307	2.312		
	Ebn at Noon		778	809	816	802	836	834	826	806	794	774	737	733		
	Edh at Noon		87	110	136	167	147	142	140	140	126	104	88	81		
(44) All-Sky Solar Radiation	RadAvg		1.54	2.32	3.45	4.76	5.97	6.49	6.70	6.02	4.45	2.88	1.79	1.36		
	RadStd		0.20	0.26	0.30	0.50	0.43	0.50	0.41	0.28	0.34	0.29	0.20	0.19		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	+2.55	+3.06	N/A	N/A	N/A
		+0.46	N/A	N/A	N/A	+0.68	+0.75	-90	-126	+91

Nomenclature: See separate page